

Q4103 Mathematical Models of Financial Derivatives

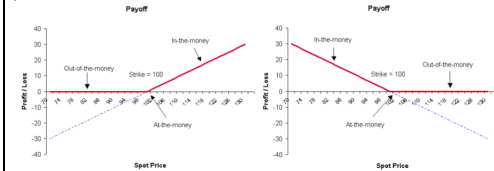
Lecture 1

Forward Contract

Definition Agreement to purchase asset at future date T at price F_0 set today

Pricing Use no-arbitrage (If $F_0 > S_0 e^{-rT}$ borrow and long stock, short forward. Vice versa)

Option Payoff (Call/Put)



Portfolio

Definition Combination of positions in different financial assets

Notation (α, β) represents α position in money market account and β position in stock. Total portfolio value at time t is $\alpha B_t + \beta S_t$.

Arbitrage Portfolio

Definition Portfolio h such that

- $V_0^h = 0$ (I start with nothing)
- $V_T^h \geq 0$ (I never lose money)
- $V_T^h \geq 0$ (I sometimes make money)

Law of One Price If $V_0^h \geq V_0^k, V_T^h \geq V_T^k$ OR if $V_0^h = V_0^k, V_T^h = V_T^k$ (in absence of arbitrage)

Binomial Model (One Period)

Consider portfolio $h = (\alpha, \beta)$, to replicate derivative, we require

- $\alpha e^{rT} + \beta S_u = \varphi(S_u)$
- $\alpha e^{rT} + \beta S_d = \varphi(S_d)$

Then the solution to the portfolio is given by

- $\alpha = e^{-rT} \frac{\varphi(S_u) - \varphi(S_d)}{S_u - S_d}$
- $\beta = \frac{1}{S_u - S_d} (\varphi(S_u) - \alpha S_u)$

And the initial price of the portfolio is given by

- $V_0^h = \alpha + \beta S_0 = e^{-rT} \left[\frac{\varphi(S_u) - \varphi(S_d)}{S_u - S_d} S_0 + \frac{S_u - e^{rT} \varphi(S_d)}{S_u - S_d} \right] = e^{-rT} \mathbb{E}^Q[\varphi(S_T)]$
- $q_u = \frac{e^{rT} - d}{u - d}, q_d = \frac{u - e^{rT}}{u - d}$ (Risk-Neutral Probabilities)

Multiperiod Binomial Model

Pricing Price of derivative is given by

$$e^{-rT} \mathbb{E}^Q[\varphi(S_T)] = e^{-rT} \sum_{k=0}^n \binom{n}{k} q_u^k q_d^{n-k} \varphi(S_0 u^k d^{n-k})$$

Cox-Ross-Rubinstein Model Discrete-time approximation of Black&Scholes Model

$$u = e^{\sigma\sqrt{\Delta t}}, d = e^{-\sigma\sqrt{\Delta t}}$$

Put-Call Parity

Definition EURO put and call with same strike price satisfy $p(K, T) = c(K, T) - S + K e^{-rT}$

Lecture 2

Wiener Process

Definition A stochastic process $W = (W_t)_{t \geq 0}$ is called a Wiener Process/Brownian Motion if

- $W_0 = 0$
- W has independent increments (if $r < s \leq t < u$, then $W_u - W_t$ and $W_s - W_r$ are independent)
- For $s < t$, $W_t - W_s \sim N(0, t - s)$
- $t \rightarrow W_t$ is continuous (with probability 1)

Properties

- Trajectories of Wiener process, $t \rightarrow W_t$ are nowhere differentiable with probability 1 (i.e. Wiener process is "spiky" no matter scale)
- If $s < t$, $\text{Cov}(W_s, W_t) = s$
- For $a > 0$, $X_t = \frac{1}{\sqrt{a}} W_{at}$ is a Wiener process

Donsker's Theorem

- The process $W_t^{(n)} := \frac{S_{[nt]} - [nt] \mu}{\sqrt{[nt]}}$ "converges" to a Wiener process

Generalized Wiener Process

Definition A generalized Wiener process is a process $X_t = at + bW_t$ where a is drift rate and b^2 is variance rate of process.

Quadratic Variation

Definition For some partition of interval $[0, T]$, quadratic variation is defined as

$$Q(W, \Delta t) = \sum_{i=0}^{n-1} (W_{t_{i+1}} - W_{t_i})^2$$

Theorem For a sequence of partitions where $\Delta t \downarrow 0$

$$\lim_{\Delta t \downarrow 0} \text{Var}[Q(W, \Delta t)] = 0$$

Stochastic Integrals

Stock price at time T is given by the Ito process

$$S_T = S_0 + \int_0^T a(t, S_t) dt + \int_0^T b(t, S_t) dW_t \quad (\text{Integral})$$

$$dS_t = a(t, S_t) dt + b(t, S_t) dW_t \quad (\text{Differential})$$

Ito's Lemma

Definition For a Stochastic Differential Equation: $dS_t = a(t, S_t) dt + b(t, S_t) dW_t$, suppose we have a function $V(t, S_t)$ that depends on S_t :

$$dV(t, S_t) = \left(\frac{\partial V}{\partial t} + a(t, S_t) \frac{\partial V}{\partial S} + \frac{1}{2} b(t, S_t)^2 \frac{\partial^2 V}{\partial S^2} \right) dt + b(t, S_t) \frac{\partial V}{\partial S} dW_t$$

Multivariate Let $S^0, S^1, \dots, S^n$ be Ito processes. Let $V(t, S^0, S^1, \dots, S^n)$ be smooth function

$$dV = \frac{\partial V}{\partial t} dt + \sum_{i=0}^n \frac{\partial V}{\partial S^i} dS^i + \frac{1}{2} \sum_{i,j=0}^n \frac{\partial^2 V}{\partial S^i \partial S^j} dS^i dS^j$$

Useful: $(dt)^2 = 0, dt \cdot dW = 0, (dB_t)^2 = (rB_t dt)^2 = 0, dS_t dB_t = 0$

Filtration

Definition Let $X := (X_t)_{t \geq 0}$ be a random variable. The set $\{F_t^X\}_{t \geq 0}$ is the filtration generated by X

Properties

- A stochastic process $Y := (Y_t)_{t \geq 0}$ is adapted to filtration $\{F_t^X\}_{t \geq 0}$ if for any t the random variable Y_t is F_t^X -measurable
- Linearity:** $\mathbb{E}[aY + bZ | F_t^X] = a\mathbb{E}[Y | F_t^X] + b\mathbb{E}[Z | F_t^X]$
- If Y is F_t^X -measurable, then $\mathbb{E}[Y | F_t^X] = Y$
- If Y is independent of F_t^X then $\mathbb{E}[Y | F_t^X] = \mathbb{E}[Y]$
- Tower Property:** If $s < t$ then $\mathbb{E}[\mathbb{E}[Y | F_t^X] | F_s] = \mathbb{E}[Y | F_s]$

Theorem Let W be a Wiener process and $G := (G_t)_{t \geq 0}$ a stochastic process such that

- $\int_0^t \mathbb{E}[G_s^2] ds < \infty$
- G is adapted to filtration $\{F_t^W\}_{t \geq 0}$

Then

- $\mathbb{E}\left[\int_0^t G_s dW_s\right] = 0$
- $\mathbb{E}\left[\left(\int_0^t G_s dW_s\right)^2\right] = \int_0^t \mathbb{E}[G_s^2] ds$ (Ito Isometry)
- $\int_0^t G_s dW_s$ is F_t^W -measurable

Lemma Let $g(s)$ be a deterministic function, then

$$\int_0^t g(s) dW_s \sim N\left(0, \int_0^t g(s)^2 ds\right)$$

Geometric Brownian Motion

Definition A Geometric Brownian Motion (GBM) is a stochastic process that satisfies the SDE

$$dS_t = S_t(\mu dt + \sigma dW_t)$$

Integral Form

$$S_t = S \cdot \exp\left\{\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma W_t\right\}$$

Risk-Free Asset The differential form for risk-free asset is given by

$$dB_t = rB_t dt$$

Chapter 3

Self-Financing Portfolio

Definition The portfolio strategy $h_t = (\alpha_t, \beta_t)$ is self-financing if

$$dV_t = \alpha_t dB_t + \beta_t dS_t$$

where

$$V_t = \alpha_t B_t + \beta_t S_t$$

Black & Scholes Equation

Definition Assume BSM, price $F(t, S_t)$ of derivative with terminal payoff $\varphi(S_T)$ satisfies

$$\frac{\partial F}{\partial t} + \frac{\partial F}{\partial S} S + \frac{1}{2} \frac{\partial^2 F}{\partial S^2} \sigma^2 S^2 - rF = 0$$

Proof

- Delta-hedging (Ito's Lemma on F , then choose Δ to remove random term)
- Replication (Build self-financing portfolio that replicates derivative)

Feynman-Kac Formula

Definition Assume $f(t, x)$ satisfies

$$\frac{\partial f}{\partial t} + a(t, x) \frac{\partial f}{\partial x} + \frac{1}{2} b^2(t, x) \frac{\partial^2 f}{\partial x^2} = 0$$

Then $f(t, x) = \mathbb{E}_{X_t=x}[h(X_T)] = \mathbb{E}[h(X_T) | F_t]$ where for $u \geq t$

$$dX_u = a(u, X_u) du + b(u, X_u) dW_u$$

Corollary Define $G(t, S) := e^{-(T-t)} F(t, S)$ then by Ito's Lemma:

$$\frac{\partial G}{\partial t} + \frac{\partial G}{\partial S} S + \frac{1}{2} \frac{\partial^2 G}{\partial S^2} \sigma^2 S^2 = 0$$

Applying Feynman-Kac Formula, we have

$$G(t, S) = \mathbb{E}_{X_t=S}[G(T, S)]$$

$$dS_t = S_t(r du + \sigma dW_u)$$

Where stock has risk-neutral dynamics.

Solving Black-Scholes-Merton Formula

Example Geometric Brownian motion under risk-neutral measure Q has dynamics

$$dS_t = S_t(r du + \sigma dW_u^Q)$$

And

$$S_T = S \cdot \exp\left\{\left(r - \frac{\sigma^2}{2}\right)(T - t) + \sigma(W_T^Q - W_t^Q)\right\}$$

We define $X = \frac{1}{\sqrt{t-T}} (W_T^Q - W_t^Q) \sim N(0, 1)$

$$S_T = S \cdot \exp\left\{\left(r - \frac{\sigma^2}{2}\right)(T - t) + \sigma\sqrt{T-t}X\right\}$$

Price of derivative is then

$$e^{-r(T-t)} \mathbb{E}_t^Q[\varphi(S_T)] = e^{-r(T-t)} \int_{-\infty}^{\infty} \left(S \cdot e^{\left(r - \frac{\sigma^2}{2}\right)(T-t) + \sigma\sqrt{T-t}x} \right) f(x) dx$$

where $f(x)$ is pdf of standard normal variable

- Substitute payoff function
- Find conditions where payoff evaluates to 0 or vice versa
- Try to fit integral into a normal distribution pdf

Chapter 4

Martingales

Definition Let $X := (X_t)_{t \geq 0}$ be a stochastic process such that

- X is adapted to filtration $\{F_t\}_{t \geq 0}$
- $\mathbb{E}[|X_t|] < \infty$ for all t

Then X is a martingale if for all $s \leq t$

- $\mathbb{E}[X_t | F_s] = X_s$
- Supermartingale** For all $s \leq t$, $\mathbb{E}[X_t | F_s] \leq X_s$
- Submartingale** For all $s \leq t$, $\mathbb{E}[X_t | F_s] \geq X_s$

Proposition Assume $\int_0^t \mathbb{E}[|x_s|^2] ds < \infty$, we let $X_t = X_0 + \int_0^t \mu_s ds + \int_0^t \sigma_s dW_s$, then X is a martingale i.f.f. $\mu_t = 0$ (i.e. drift term = 0)

Martingale Representation Theorem

Definition Let W_t be a Wiener process and $F := (F_t)_{t \geq 0}$ the filtration generated by W_t . Let $M := (M_t)_{t \geq 0}$ be a martingale with respect to F . Then there exists a process h adapted to F such that

$$M_t = M_0 + \int_0^t h_s dW_s$$

Girsanov Theorem (Simplified)

Definition Let W_t^P be a Wiener process under measure P . For some constant k , define the process

$$L_t = \exp\left\{kW_t^P - \frac{1}{2}k^2 t\right\}$$

Which has dynamics $dL_t = kL_t dW_t^P$. Define Q on F_t by

$$L_T = \frac{dQ}{dP}$$

Then

$$W_t^Q = W_t^P - kt$$

Is a Wiener process under Q

Intuition We find some k so that the process satisfies the given dynamics. Then we can directly convert the old Wiener process into a Wiener process under a new measure.

Girsanov Theorem (General)

Definition Let W_t^P be a Wiener process under measure P . Let k_t be an adapted process and define

$$dL_t = k_t L_t dW_t^P$$

Then

$$dW_t^Q = dW_t^P - k_t dt$$

Where W_t^Q is a Wiener process under Q

Novikov Condition

Definition If $\mathbb{E}^P\left[\exp\left\{\frac{1}{2}\int_0^T |k_t|^2 dt\right\}\right] < \infty$ then $\mathbb{E}^P[L_T] = 1$

Greeks

- Delta**
 $\Delta = \frac{\partial P}{\partial S}(t, S_t)$
- Gamma**
 $\Gamma = \frac{\partial^2 P}{\partial S^2}(t, S_t)$
- Theta**
 $\Theta = \frac{\partial P}{\partial t}(t, S_t)$
- Vega**
 $v = \frac{\partial P}{\partial \sigma}(t, S_t)$

Gamma-Neutral Portfolio

For every unit of financial derivative F , we also trade $y = \frac{r_F}{r_K}$ units of another derivative G .

Black & Scholes with Continuous Dividend Yield

$$\frac{\partial F}{\partial t} + (r - q) \frac{\partial F}{\partial S} S + \frac{1}{2} \frac{\partial^2 F}{\partial S^2} \sigma^2 S^2 - rF = 0$$

Chapter 5

Change of Numeraire

Bond Numeraire Price of stock renormalized by value of the bond, $\frac{S_t}{B_t}$ is a martingale under Q

$$\frac{V_t}{B_t} = \mathbb{E}_t^Q\left[\frac{V_T}{B_T}\right]$$

Risk-Neutral Valuation Formula The price at time t of a financial derivative with payoff X paid at terminal time T is

$$V_t = e^{-r(T-t)} \mathbb{E}[X | F_t] = B_t \mathbb{E}^Q\left[\frac{X}{B_T} \middle| F_t\right]$$

Choice of Numeraire Any positive financial asset can be chosen as numeraire

Change of Numeraire Formula Let N_t be a numeraire. There exists a measure Q^N such that every European-style derivative with payoff X at T has price V_t that satisfies

$$\frac{V_t}{N_t} = \mathbb{E}_t^{Q^N}\left[\frac{X}{N_T}\right]$$

American Options

Option Value Bounds Lower/Upper case for European/American

- $S_t \geq C(S_t, t) \geq c(S_t, t) \geq (S_t - K e^{-r(T-t)})^+ \geq (S_t - K)^+$
- $K \geq P(S_t, t) > p(S_t, t) \geq (K e^{-r(T-t)} - S_t)^+ \geq (K - S_t)^+$

Note: American put can be worth strictly more than European put by considering random variable $\tau = \min\{t \geq 0: S_t \leq K - K e^{-r(T-t)}\}$ and exercising at $\min(\tau, T)$

No Dividend When there are no dividends and interest rates are positive

- $C(S_t, t) \leq c(S_t, t)$ (Early Exercise feature for American call is worthless)

Optimal Exercise Boundary

Problem Find some value S^* which separates early exercise and hold region.

- Early Exercise Region:** $F(S, t) = \varphi(S, t)$ (Value equals payoff)
- Holding:** $F(S, t) > \varphi(S, t)$ (Holding is better than early exercise payoff)

Exercise and Continuation Regions Inequality signs depend on instrument

- $E = \{(S, t) \in D: S \leq (\geq) S^*\}$
- $H = \{(S, t) \in D: S > (<) S^*\}$

Smooth Pasting Condition

$$\frac{\partial P}{\partial S} = -1 \text{ at } S = S_p(t)$$

$$\frac{\partial C}{\partial S} = 1 \text{ at } S = S_c(t)$$

Near Complementarity Problem

Problem Find inequalities that can be combined into one expression

Holding Region:

$$-L_{BS}(F) := -\frac{\partial F}{\partial t} - \frac{\partial F}{\partial S} S - \frac{1}{2} \frac{\partial^2 F}{\partial S^2} \sigma^2 S^2 + rF = 0$$

Early Exercise:

$$F(S, t) - \varphi(S, t) = 0$$

$$-L_{BS}(F) := -\frac{\partial F}{\partial t} - \frac{\partial F}{\partial S} S - \frac{1}{2} \frac{\partial^2 F}{\partial S^2} \sigma^2 S^2 + rF > 0$$

Combined:

$\min[-L_{BS}(F), F - \varphi] = 0$
Where domain is $D = [0, T] \times (0, +\infty)$
Free Boundary Problem
American Put/Call
If $S > S_p(t)$, then $P(S, t) (C(S, t))$ solves the Black-Scholes PDE
$L_{BS}(P) = \frac{\partial P}{\partial t} + r \frac{\partial P}{\partial S} S + \frac{1}{2} \frac{\partial^2 P}{\partial S^2} \sigma^2 S^2 - rP = 0$ (Replace P with C for Call)
$P(S_p(t), t) = K - S_p(t) = (S_c(t) - K)$

When barrier is hit, option is worthless: $V(t, B) = 0, \quad \forall t \in [0, T]$
Solution domain for lower barrier is $[0, T] \times [B, +\infty)$
Final condition is payoff of corresponding vanilla option $V(T, S) = \varphi(S)$

Chapter 7

Asian Options

$A_T = \begin{cases} \frac{1}{n} \sum_{i=1}^n S_{t_i}, & \text{discretely sampled arithmetic} \\ \frac{1}{T} \int_0^T S_t dt, & \text{continuously sampled arithmetic} \\ \exp\left(\frac{1}{n} \sum_{i=1}^n \ln S_{t_i}\right) = (S_{t_1} \dots S_{t_n})^{\frac{1}{n}}, & \text{discretely sampled geometric} \\ \exp\left(\frac{1}{T} \int_0^T \log S_t dt\right), & \text{continuously sampled geometric} \end{cases}$
Types of strike: fixed strike or floating strike Only consider continuously sampled Asian options. Arithmetic Asian: no closed form analytic solution

Pricing Geometric Asian Option (Fixed Strike Call Example)

Step 1: Rewrite geometric average $G_t = \exp\left\{\left(\frac{1}{t} \int_0^t \ln(S_u) du\right)\right\} = \exp\{X_t\}$ $\varphi(S_T, G_t) = (e^{X_T} - K)^+$
Step 2: Price of option at time t is given by $e^{-r(T-t)} E_t^Q[(G_T - K)^+]$
Step 3: Rewrite X_t into F_t -measurable & independent parts $\ln(S_u) - \ln(S_t) = \left(r - \frac{\sigma^2}{2}\right)(u - t) + \sigma(W_u - W_t)$ $X_T = \frac{1}{T} \int_0^T \ln(S_u) du = \frac{1}{T} \int_0^t \ln(S_u) du + \frac{1}{T} \int_t^T \ln(S_u) du$ $= \frac{t}{T} X_t + \frac{1}{T} \int_t^T \ln(S_u) + \left(r - \frac{\sigma^2}{2}\right)(u - t) + \sigma(W_u - W_t) du$ $= \frac{t}{T} X_t + \frac{T-t}{T} \ln(S_t) + \frac{1}{T} \int_t^T \left(r - \frac{\sigma^2}{2}\right)(u - t) du + \frac{\sigma}{T} \int_t^T (W_u - W_t) du$ $= \frac{t}{T} X_t + \frac{T-t}{T} \ln(S_t) + \frac{1}{T} \left(r - \frac{\sigma^2}{2}\right) \frac{(T-t)^2}{2T}$ $+ \frac{\sigma}{T} \int_t^T (W_u - W_t) du$
Step 4: Use "Integrating Wiener Processes" result to get *Note: $(Y \sim N(0, 1))$ $\tilde{\mu} = \left(r - \frac{\sigma^2}{2}\right) \frac{(T-t)^2}{2T}, \tilde{\sigma} = \frac{\sigma}{T} \frac{(T-t)^{3/2}}{3^{1/2}}$ $G_T = e^{X_T} = e^{\frac{T-t}{T} \ln(S_t)} \exp\left[\tilde{\mu} + \tilde{\sigma} Y\right] = G_t^{1/T} S_t^{(T-t)/T} \exp\left[\tilde{\mu} + \tilde{\sigma} Y\right]$
Step 5: Substitute back into pricing function $e^{-r(T-t)} E_t^Q[(G_T - K)^+] = e^{-r(T-t)} E_t^Q\left[\left(G_t^{1/T} S_t^{(T-t)/T} \exp\left[\tilde{\mu} + \tilde{\sigma} Y\right] - K\right)^+\right]$
Step 6: Break into case where $G_T > K$ to remove outer function to get $e^{-r(T-t)} \left(G_t^{1/T} S_t^{(T-t)/T} e^{\frac{1}{2} \tilde{\sigma}^2} N(d_1) - K N(d_2)\right), d_2 = \frac{1}{\tilde{\sigma}} \left(\ln G_t + \frac{T-t}{T} \ln S_t - \ln K + \tilde{\mu}\right), d_1 = d_2 + \tilde{\sigma}$

PDE Framework

Goal: Find equivalent B&S for path dependent options. Price of option given by $F(S_t, t, I_t)$ $I_t = \int_0^t f(S_u, u) du, \quad dI_t = f(S_t, t) dt$
Step 1: Apply multivariate Ito's lemma $(dI_t \cdot dI_t = 0, dS_t \cdot dI_t = 0)$ $dF_t = \frac{\partial F}{\partial t} dt + \frac{\partial F}{\partial S} dS_t + \frac{\partial F}{\partial I} dI_t + \frac{1}{2} \frac{\partial^2 F}{\partial S^2} (dS_t)^2 + \frac{1}{2} \frac{\partial^2 F}{\partial I^2} (dI_t)^2 + \frac{\partial^2 F}{\partial S \partial I} (dS_t)(dI_t)$ $= \left(\frac{\partial F}{\partial t} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 F}{\partial S^2}\right) dt + \frac{\partial F}{\partial S} dS_t + \frac{\partial F}{\partial I} dI_t$
Step 2: Build self-financing and risk-free portfolio of option and stocks $V_t = \gamma_t F_t + \beta_t S_t$ $dV_t = \gamma_t dF_t + \beta_t dS_t = \gamma_t \left(\frac{\partial F}{\partial t} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 F}{\partial S^2} + \frac{\partial F}{\partial I} f(S_t, t, I_t)\right) dt + \gamma_t \frac{\partial F}{\partial S} dS_t + \beta_t dS_t$
Step 3: Choose $\beta_t = -\frac{\partial F}{\partial S} \gamma_t$ $dV_t = \gamma_t \left(\frac{\partial F}{\partial t} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 F}{\partial S^2} + \frac{\partial F}{\partial I} f(S_t, t, I_t)\right) dt = rV_t dt$ $V_t = \gamma_t F_t + \beta_t S_t = \gamma_t F_t - \gamma_t \frac{\partial F}{\partial S} S_t \quad \gamma_t = \frac{V_t}{F_t - \frac{\partial F}{\partial S} S_t}$ $\therefore \frac{\partial F}{\partial t} + rS_t \frac{\partial F}{\partial S} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 F}{\partial S^2} + \frac{\partial F}{\partial I} f(S_t, t) - rF = 0$

Arithmetic Asian: Payoff depends on $A_T = \frac{1}{T} \int_0^T S_u du$ Choose $f(S, t)$ such that $I_t = \int_0^t f(S_u, u) du$ contain all info on A_T at T
Choice 1: $f(S_t, t) = S_t$ $dI_t = S_t dt, A_T = \frac{I_T}{T}$ It follows that the PDE for Arithmetic Asian option is $\frac{\partial F}{\partial t} + rS \frac{\partial F}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 F}{\partial S^2} + S \frac{\partial F}{\partial I} - rF = 0$ Terminal condition is $F(T, S, I) = \begin{cases} (kI_T - S)^+, & \text{floating put} \\ (S - kI_T)^+, & \text{floating call} \\ \left(K - \frac{I_T}{T}\right)^+, & \text{fixed put} \\ \left(\frac{I_T}{T} - K\right)^+, & \text{fixed call} \end{cases}$
Choice 2: f such that $I_t = A_t$ $dA_t = \frac{S_t - A_t}{T} dt$ $f(S_t, A_t, t) = \frac{1}{T}(S_t - A_t)$ It follows that the PDE for Arithmetic Asian option is $\frac{\partial F}{\partial t} + rS_t \frac{\partial F}{\partial S} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 F}{\partial S^2} + \frac{S - A}{T} \frac{\partial F}{\partial A} - rF = 0$ Terminal condition is $F(T, S, A) = \begin{cases} (kA - S)^+, & \text{floating put} \\ (S - kA)^+, & \text{floating call} \\ (K - A)^+, & \text{fixed put} \\ (A - K)^+, & \text{fixed call} \end{cases}$

Roger-Shi method: Transformation to reduce dim of the PDE for Arith Asian

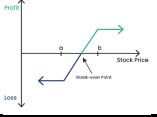
$E_t^Q[(A_T - K)^+] = E_t^Q\left[\left(\frac{1}{T} \int_0^T S_u du - K\right)^+\right] = E_t^Q\left[\left(\frac{1}{T} \int_0^T S_u du - \left(K - \frac{1}{T} \int_0^t S_u du\right)\right)^+\right]$ $= S_t E_t^Q\left[\left(\frac{t}{T} \int_t^T \frac{S_u}{S_t} du - X_t\right)^+\right] \quad \text{where } X_t = \frac{K - \frac{1}{T} \int_0^t S_u du}{S_t} = \frac{K - \frac{1}{T} S_t}{S_t}$ For $u > t, \frac{S_u}{S_t} = \exp\left\{\left(r - \frac{\sigma^2}{2}\right)(u - t) + \sigma(W_u - W_t)\right\}$ indep of history asset price, $E_t^Q\left[\left(\frac{t}{T} \int_t^T \frac{S_u}{S_t} du - X_t\right)^+\right]$ fun of X_t and t only
Define $H(X_t, t) = e^{-r(T-t)} E_t^Q\left[\left(\frac{1}{T} \int_t^T \frac{S_u}{S_t} du - X_t\right)^+\right]$ Price at time t of arithmetic Asian call option is $F(t, S_t, I_t) = S_t H(X_t, t)$ Recall that $\frac{\partial F}{\partial t} + rS \frac{\partial F}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 F}{\partial S^2} + S \frac{\partial F}{\partial I} - rF = 0$ $X_t = \frac{K - \frac{1}{T} S_t}{S_t}, \quad \frac{\partial X_t}{\partial S} = -\frac{X}{S}$ $\frac{\partial F}{\partial t} = S \frac{\partial H}{\partial t}, \quad \frac{\partial F}{\partial S} = H - X \frac{\partial H}{\partial X}, \quad \frac{\partial^2 F}{\partial S^2} = \frac{X^2}{S} \frac{\partial^2 H}{\partial X^2}, \quad \frac{\partial F}{\partial I} = -\frac{1}{S} \frac{\partial H}{\partial X}$ Sub into PDE: $\frac{\partial H}{\partial t} - \frac{\partial H}{\partial X} \left(\frac{1}{t} + rX\right) + \frac{1}{2} \sigma^2 X^2 \frac{\partial^2 H}{\partial X^2} = 0$ Where $X \in (-\infty, +\infty)$ and $t \in [0, T)$ The terminal condition becomes $F(S, I, T) = \left(\frac{I}{T} - K\right)^+ = S_t \left(-\frac{K - \frac{1}{T} S_t}{S_t}\right)^+ = S(-X)^+, \quad H(X, T) = (-X)^+$
The case $X \leq 0$: Find explicit solution. If $s > t$, then $I_s > I_t$. If $X_t \leq 0$: $S_t X_s = K - \frac{I_s}{T} \leq K - \frac{I_t}{T} + S_t X_t \leq 0$ In particular, if $X_t \leq 0$, $\frac{1}{T} \int_t^T \frac{S_u}{S_t} du - X_s \geq 0, \forall s \geq t$ $H(S_t, t) = e^{-r(T-t)} E_t^Q\left[\left(\frac{1}{T} \int_t^T \frac{S_u}{S_t} du - X_t\right)^+\right]$ $= e^{-r(T-t)} E_t^Q\left[\frac{1}{T} \int_t^T \frac{S_u}{S_t} du - X_t\right] = e^{-r(T-t)} \frac{1}{T} \int_t^T E_t^Q\left[\frac{S_u}{S_t}\right] du - e^{-r(T-t)} X_t$ $= e^{-r(T-t)} \frac{1}{T} \int_t^T E_t^Q\left[\exp\left\{\left(r - \frac{\sigma^2}{2}\right)(u - t) + \sigma(W_u - W_t)\right\}\right] du - e^{-r(T-t)} X_t$ $= e^{-r(T-t)} \frac{1}{T} \int_t^T e^{(u-t)} du - e^{-r(T-t)} X_t$ $= e^{-r(T-t)} \frac{1}{T} \left[\frac{e^{(u-t)}}{r}\right]_t^T = e^{-r(T-t)} X_t = \frac{1 - e^{-r(T-t)}}{rT} e^{-r(T-t)} X_t$ $\therefore F(t, S, I) = SH(X, t) = S \left(\frac{1 - e^{-(r-t)}}{rT} - e^{-(r-t)} X_t\right) = S \frac{1 - e^{-r(T-t)}}{rT} = e^{-r(T-t)} \left(\frac{I}{T} - K\right)$
The case $X > 0$: No explicit solution, $H(X, t)$ is found numerically via $\frac{\partial H}{\partial t} - \frac{\partial H}{\partial X} \left(\frac{1}{t} + rX\right) + \frac{1}{2} \sigma^2 X^2 \frac{\partial^2 H}{\partial X^2} = 0$ Restrict domain to $(0, +\infty)$ $\begin{cases} \frac{\partial H}{\partial t} - \frac{\partial H}{\partial X} \left(\frac{1}{t} + rX\right) + \frac{1}{2} \sigma^2 X^2 \frac{\partial^2 H}{\partial X^2} = 0 \\ H(0, t) = \frac{1 - e^{-r(T-t)}}{rT} \\ H(\infty, t) = 0 \\ H(X, T) = 0, \quad X \in (0, +\infty) \end{cases}$

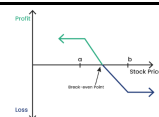
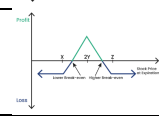
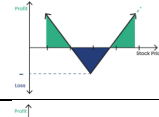
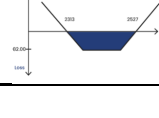
Useful Results

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Variance $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$
Covariance $\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$
Product of Independent Random Variables $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$
n-th Moment of Standard Gaussian Let $X \sim N(0, 1)$, then $\mathbb{E}[X^n] = \begin{cases} 0, & n \text{ is odd} \\ (n-1)(n-3) \dots (3)(1), & n \text{ is even} \end{cases}$
Rules for differential terms (dW_t must be WIENER process) $(dt)^2 = 0$ $dt \cdot dW_t = 0$ $(dW_t)^2 = dt$
Expectation of exponential of normal r.v. Suppose $X \sim N(a, b^2)$ we have $\mathbb{E}[e^X] = e^{a + \frac{1}{2}b^2}$
Normal Random Variable PDF $f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x-\mu}{\sigma}\right)^2}$
European Call/Put Option Solution $c = S_t e^{-q(T-t)} N(d_1) - K e^{-r(T-t)} N(d_2)$ $d = \frac{\ln\left(\frac{S_t}{K}\right) + \left(r - q + \frac{1}{2} \sigma^2\right)(T - t)}{\sigma \sqrt{T - t}}$ $d_1 = \frac{d_2 + \sigma \sqrt{T - t}}{\sigma \sqrt{T - t}}$
Integrating Wiener Processes Let $(W_t)_{t \geq 0}$ be a Wiener process. Then $\int_t^T (W_u - W_t) du \sim N\left(0, \frac{(T-t)^3}{3}\right)$
Multivariate Chain Rule If $e(S, t) = X(t)f(S, X)$ $\frac{\partial e}{\partial t} = \frac{dX}{dt} f(S, X) + \frac{\partial f}{\partial X} \frac{dX}{dt}$
Random Equalities $S_t N'(d_1) = K e^{-r(T-t)} N'(d_2)$

Option Strategies

Bull Spread (Long 1 call strike K_1 , Short 1 call strike $K_2, K_1 < K_2$) Alternatively, long 1 put K_1 , short put K_2


Bear Spread (Short 1 unit put strike K_1 , Long 1 put strike $K_2, K_1 < K_2$) Alternatively, short 1 call K_1 , long 1 call K_2

Butterfly Spread (Long 1 unit call K_1 , Long 1 unit call K_3 , Short 2 unit call K_2) Alternatively, long 1 unit K_1 put, long 1 unit K_3 put, short 2 unit K_2 put

Long Straddle (Long 1 unit call with strike K and Long 1 unit put with strike K) Can also take short position to reflect payoff about x-axis

Long Strangle (Long 1 unit call with strike K_2 , Long 1 unit put with strike K_1) Can also invert positions to invert payoff about x-axis


Problem Solving Techniques

Proving process is martingale 1. Apply Ito's Lemma 2. If dt term is 0, can conclude martingale
Finding Expectation of Max/Min of Wiener 1. Find bounds where $P(M_T \leq x) = 0$ 2. Apply Reflection principle where applicable 3. Get distribution $f(x)$ by taking derivative w.r.t x 4. Expectation given by $\int xf(x)dx$
Pricing Options with Weird Payoff - Option 1: Write as risk-neutral pricing formula, with indicator function in expectation function to take care of any $(\cdot)^+$ functions - Option 2: Use change of numeraire if there are S_T terms that can be factorized. Then substitute usual risk-neutral pricing, but replace the r with drift term of new process. - Option 3: Use Ito's lemma by rewriting $Y_t = f(S_t)$ where $f(S_t)$ is the S_t term in the payoff function. Then try to rewrite payoff with Y_t into result we already know, apply risk-neutral pricing to directly substitute formula, replacing the r with drift term of new process.